

PAPER_856: Higgs Field UH Vacuum Excitation via UQFF Pseudo-Monopole Density

Author: Daniel T. Murphy – Star Magic / UQFF Framework **Date:** 2026-04-04 **Session:** 199 **Source:** grok_share_589f6949-6fe9.txt (2404 lines, May 05 – Aug 07, 2025) **Calculator:** HiggsVacuumUHExcitationKHiggsUQFFCalc (CP4 #440) **CVW:** v2.0.0 compliant

Abstract

We present a UQFF derivation of the Higgs field energy density $U_H(t,n)$ from pseudo-monopole vacuum excitation. The equation $U_H = \lambda_H \cdot \rho_{vac} \cdot \omega_H \cdot \exp(-[SSq]n/26) \cdot \exp(-(\pi - t)) \cdot (1 + f_{quasi})$ yields $U_H \sim 1.539e-32 \text{ J/m}^3$ at $n=1, t=0$, corresponding to $E_H \sim 96.25 \text{ eV}$. The scaling factor $k_{Higgs} = 125 \text{ GeV} / E_H \sim 1.79e18$ bridges the UQFF vacuum scale to the observed Higgs boson mass of 125 GeV, identifying the multiplicative vacuum-to-particle amplification mechanism.

1. Core Equations

- $U_H(t, n) = \lambda_H \cdot \rho_{vac}, [UA']:[SCm](n,t) \cdot \omega_H(t) \cdot \exp(-[SSq]n/26) \cdot \exp(-(\pi - t)) \cdot (1 + f_{quasi})$
 - $m_H = U_H / c^2 \sim 1.711e-49 \text{ kg}$
 - $E_H = m_H \cdot c^2 \sim 1.54e-41 \text{ J} \sim 96.25 \text{ eV}$
 - $k_{Higgs} = 125 \text{ GeV} \cdot 1.602e-19 / E_H \sim 1.79e18$
 - $\omega_H = \omega_c \sim 1.585e-8 \text{ rad/s}, \lambda_H = 1.0, f_{quasi} = 0.01$
-

2. UQFF Integration

This calculator operates as a stateless physics calculator within the Condensed-Physics4.py (Phase 4) IPC chain. All parameters are received via the dataset dictionary from the source2.cpp principal GUI pipeline. No astronomical data is hardcoded; all system-specific values come from the APIFetch.py -> bodies_*.csv data flow.

3. Source Data

- **File:** grok_share_589f6949-6fe9.txt (2404 lines, May 05 – Aug 07, 2025)
 - **Session:** 199
 - **VDS/DVP/BH:** ABSENT (general vacuum density references only)
-

References

1. Murphy, D.T. – Star Magic UQFF Framework (2024-2026)
2. Srivastava, Y.N., Widom, A., Larsen, L. – Electroweak neutron production (LENR)
3. Kepler Mission DR25 – 4,034 candidates, 2,335 confirmed planets
4. Hubble Heritage Team / A. Nota (ESA/STScI) – Westerlund 2 / NGC 346 imaging
5. UQFF Calibration: $\kappa=0.0005/\text{day}$, $[SSq]=0.57$, $\beta_i \sim 0.603$